Lab4(EXP 6)

1. Concurrent Tasks Using Goroutines and Channels

1. Write a Go program that launches at least three goroutines concurrently to perform independent tasks (e.g., computing squares, cubes, and Fibonacci numbers).
2. Use a channel of the appropriate type to collect and print results in main().
3. Add print statements showing goroutine start and completion to demonstrate concurrent execution order.



```
> Users > Jigyasa singh > Downloads > exp6.go > ...
package main

import (
    "fmt"
    "sync"
)

func calculateSquares(ch chan string, wg *sync.WaitGroup) {
    defer wg.Done()

    fmt.Println("Square Goroutine Started")

    for i := 1; i <= 5; i++ {
        result := i * i
        ch <- fmt.Sprintf("Square of %d = %d", i, result)
    }

    fmt.Println("Square Goroutine Completed")
}

func calculateCubes(ch chan string, wg *sync.WaitGroup) {
    defer wg.Done()

    fmt.Println("Cube Goroutine Started")

    for i := 1; i <= 5; i++ {
        result := i * i * i
        ch <- fmt.Sprintf("Cube of %d = %d", i, result)
    }

    fmt.Println("Cube Goroutine Completed")
}

func calculateFibonacci(ch chan string, wg *sync.WaitGroup) {
    defer wg.Done()

    fmt.Println("Fibonacci Goroutine Started")

    a, b := 0, 1

    for i := 1; i <= 5; i++ {
```

```go
func calculateFibonacci(ch chan string, wg *sync.WaitGroup) {
    defer wg.Done()

    fmt.Println("Fibonacci Goroutine Started")

    a, b := 0, 1

    for i := 1; i <= 5; i++ {
        ch <- fmt.Sprintf("Fibonacci number %d = %d", i, a)
        a, b = b, a+b
    }

    fmt.Println("Fibonacci Goroutine Completed")
}

func main() {
    ch := make(chan string)
    var wg sync.WaitGroup

    wg.Add(3)

    go calculateSquares(ch, &wg)
    go calculateCubes(ch, &wg)
    go calculateFibonacci(ch, &wg)

    go func() {
        wg.Wait()
        close(ch)
    }()

    fmt.Println("Main function collecting results:")

    for result := range ch {
        fmt.Println(result)
    }

    fmt.Println("All goroutines completed.")
}
```

```
PS C:\Users\Jigyasa singh> go run "c:\Users\Jigyasa singh\Downloads\exp6.go"
Main function collecting results:
Cube Goroutine Started
Fibonacci Goroutine Started
Square Goroutine Started
Cube of 1 = 1
Cube of 2 = 8
Fibonacci number 1 = 0
Square of 1 = 1
Cube of 3 = 27
Fibonacci number 2 = 1
Square of 2 = 4
Cube of 4 = 64
Fibonacci number 3 = 1
Square of 3 = 9
Cube of 5 = 125
Fibonacci number 4 = 2
Square of 4 = 16
Fibonacci number 5 = 3
Square of 5 = 25
Fibonacci Goroutine Completed
Cube Goroutine Completed
Square Goroutine Completed
```